

MAT41-5

Applied Estimation & Data Modeling

Density estimation, trees, ensembles, and model selection

The mathematical structure of tree-based and ensemble methods, plus practical density estimation.

Albert School — 26 hours

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1 Kernel density estimation

We observe an i.i.d. sample $x_1, \dots, x_n$ from an unknown density f and wish to estimate f without assuming a parametric form.

1.1 Histograms as piecewise-constant density estimates

Partition the real line into bins of width h with origin x_0 , the bin $B_k = [x_0 + kh, x_0 + (k+1)h)$. The histogram estimator is

$$\hat{f}_{\text{hist}}(x) = \frac{\text{count of } x_i \in B_k}{nh} \quad \text{for } x \in B_k.$$

The division by nh makes the estimate integrate to 1, so it is a genuine density estimate. It is constant on each bin, hence piecewise constant. Its two defects are the dependence on the bin origin x_0 and its discontinuity at bin edges.

1.2 KDE as a smoothed histogram

A kernel K is a function with $\int K(u) du = 1$, $K(u) = K(-u)$, and $\int u^2 K(u) du < \infty$. The kernel density estimator with bandwidth $h > 0$ is

$$\hat{f}_h(x) = \frac{1}{nh} \sum_{i=1}^n K\left(\frac{x - x_i}{h}\right).$$

Each observation contributes a bump of total mass $\frac{1}{n}$ centred at x_i ; the estimate is their sum. Choosing K to be the uniform density on $[-\frac{1}{2}, \frac{1}{2}]$ recovers a histogram-like estimate that is centred at x rather than tied to a fixed bin grid, which is the sense in which KDE is a smoothed histogram.

Definition 1 Common kernels: the **Gaussian** $K(u) = (2\pi)^{-\frac{1}{2}} e^{-\frac{u^2}{2}}$, the **Epanechnikov** $K(u) = \frac{3}{4}(1 - u^2)$ for $|u| \leq 1$, the **uniform** $K(u) = \frac{1}{2}$ for $|u| \leq 1$, and the **triangular** $K(u) = 1 - |u|$ for $|u| \leq 1$.

1.3 Kernel choice

Different kernels yield estimators of similar quality; the Epanechnikov kernel minimizes asymptotic mean integrated squared error among non-negative kernels, but the efficiency loss of other smooth kernels is small. The choice of kernel matters far less than the choice of bandwidth.

1.4 Bias–variance at the function level

For a fixed x , a Taylor expansion of the expectation of $\hat{f}_h(x)$ gives, writing $\mu_2(K) = \int u^2 K(u) du$ and $R(K) = \int K(u)^2 du$,

$$\text{bias}(\hat{f}_h(x)) \approx \frac{1}{2} h^2 \mu_2(K) f''(x), \quad \text{Var}(\hat{f}_h(x)) \approx \frac{f(x) R(K)}{nh}.$$

Increasing h raises the (squared) bias as h^4 and lowers the variance as $\frac{1}{nh}$. The mean integrated squared error $\text{MISE} = \int [\text{bias}^2 + \text{Var}] dx$ is minimized by

$$h^* = \left(\frac{R(K)}{\mu_2(K)^2 \int f''(x)^2 dx} \right)^{\frac{1}{5}} n^{-\frac{1}{5}},$$

so the optimal bandwidth shrinks at rate $n^{-\frac{1}{5}}$ and the best attainable MISE decreases at rate $n^{-\frac{4}{5}}$.

1.5 Bandwidth selection

Definition 2 (Silverman's rule) For a Gaussian kernel and reference normal density, the plug-in optimal bandwidth is

$$\hat{h} = 0.9 \min\left(\hat{\sigma}, \frac{\text{IQR}}{1.34}\right) n^{-\frac{1}{5}},$$

where $\hat{\sigma}$ is the sample standard deviation and IQR the interquartile range.

Definition 3 (Cross-validation bandwidth) Choose h to minimize the least-squares CV criterion

$$\text{CV}(h) = \int \hat{f}_h(x)^2 dx - \frac{2}{n} \sum_{i=1}^n \hat{f}_{h,-i}(x_i),$$

where $\hat{f}_{h,-i}$ is the estimator built from the sample with x_i removed. This is an unbiased estimate (up to a constant) of $\int (\hat{f}_h - f)^2$.

1.6 Visualization and comparison

KDE outperforms the histogram when the underlying density is smooth: the histogram is piecewise constant and origin-dependent, whereas KDE is continuous and origin-free, achieving the same $n^{-\frac{4}{5}}$ MISE rate with a smaller constant. Histograms remain adequate for quick exploratory summaries and for genuinely discontinuous densities.

2 Decision trees

A decision tree recursively partitions the feature space into regions and assigns a constant prediction to each region.

2.1 Entropy as a splitting criterion

For a node containing a fraction p of positive examples (binary classification), the Shannon entropy is, self-contained,

$$H(p) = -p \log p - (1-p) \log(1-p),$$

with the convention $0 \log 0 = 0$. It measures node impurity: $H = 0$ when $p \in \{0, 1\}$ (pure node) and H is maximal at $p = \frac{1}{2}$. For C classes with proportions $p_1, \dots, p_C$, $H = -\sum_{c=1}^C p_c \log p_c$.

2.2 Gini impurity

An alternative impurity measure is the Gini index

$$G(p_1, \dots, p_C) = \sum_{c=1}^C p_c(1-p_c) = 1 - \sum_{c=1}^C p_c^2,$$

the probability that two independently drawn labels from the node disagree. Like entropy it is 0 for a pure node and maximal at the uniform distribution; it avoids the logarithm and is cheaper to compute.

2.3 Information gain

A split s sends a fraction w_L of a node's examples to a left child and $w_R = 1 - w_L$ to a right child. The information gain is the impurity decrease

$$\text{IG}(s) = H_{\text{parent}} - (w_L H_{\text{left}} + w_R H_{\text{right}}),$$

using H (or G) as the impurity. The chosen split maximizes IG.

2.4 Recursive partitioning

Starting from the root, at each node enumerate candidate splits (for a numeric feature, thresholds between consecutive sorted values; for a categorical feature, subsets), pick the split with maximal information gain, and recurse on each child. Stopping criteria: a node is made a leaf when it is pure, when its size falls below a minimum, when the maximum depth is reached, or when no split yields positive gain.

2.5 Pruning via minimum description length

A fully grown tree overfits. Pruning trades fit against size. Under the minimum description length (MDL) principle, the cost of a tree T is the number of bits to encode the model plus the bits to encode the data given the model:

$$L(T) = L(\text{tree}(T)) + L(\text{data} \mid T).$$

$L(\text{data} \mid T)$ decreases as the tree fits the training data better, while $L(\text{tree}(T))$ grows with the number of nodes. Pruning removes subtrees whenever doing so lowers the total $L(T)$, i.e. whenever the bits saved in describing a smaller tree exceed the extra bits needed to describe its errors.

3 Ensemble methods: bagging

3.1 Bootstrap aggregation

Given a training set of size n , draw B bootstrap samples (each n points sampled with replacement), fit a model $\hat{g}_b$ on each, and average:

$$\hat{g}_{\text{bag}}(x) = \frac{1}{B} \sum_{b=1}^B \hat{g}_b(x)$$

for regression, or majority vote for classification.

3.2 Variance reduction mechanism

If the individual models each have variance σ^2 and pairwise correlation ρ , the variance of the average of B of them is

$$\text{Var}(\hat{g}_{\text{bag}}) = \rho\sigma^2 + \frac{1-\rho}{B}\sigma^2.$$

Averaging drives the second term to 0 as $B \rightarrow \infty$, leaving $\rho\sigma^2$. Because each $\hat{g}_b$ is fit on a resample of the same distribution, averaging leaves the bias essentially unchanged while reducing variance. Bagging therefore helps most for low-bias, high-variance base learners such as deep trees.

3.3 Out-of-bag estimation

In a bootstrap sample, the probability that a given observation is omitted is $(1 - \frac{1}{n})^n \rightarrow e^{-1} \approx 0.368$. Thus each point is out-of-bag for about a third of the trees. The OOB prediction for x_i averages only the trees for which x_i was out-of-bag; the OOB error aggregates these and estimates test error without a separate validation set.

4 Random forests

4.1 Decorrelation via feature subsampling

A random forest is bagging of decision trees with one addition: at each split, only a random subset of m of the d features is considered as split candidates (typically $m \approx \sqrt{d}$ for classification). This prevents every tree from splitting first on the same dominant feature.

4.2 Why variance reduction improves

Recall $\text{Var}(\hat{g}_{\text{bag}}) = \rho\sigma^2 + \frac{1-\rho}{B}\sigma^2$. As $B \rightarrow \infty$ this tends to $\rho\sigma^2$, so the floor on variance is set by the correlation ρ between trees. Feature subsampling lowers ρ by forcing trees to differ in their splits, pushing the floor $\rho\sigma^2$ down. This is the random forest improvement over plain bagging.

4.3 Variable importance

Definition 4 (Impurity importance) The importance of a feature is the total decrease in node impurity (weighted by the fraction of samples reaching the node) summed over all splits on that feature, averaged over all trees.

Definition 5 (Permutation importance) For each feature, randomly permute its values in the OOB samples and measure the resulting increase in OOB error. A large increase indicates a feature on which predictions depend strongly.

Importance scores rank features by their predictive contribution; a near-zero score indicates a feature the forest does not rely on.

5 Ensemble methods: boosting

5.1 Additive model formulation

Boosting builds an additive model in a stagewise manner:

$$F_M(x) = \sum_{m=1}^M \nu h_m(x),$$

where each h_m is a base learner (a small tree) and ν is the learning rate. At stage m the current model is F_{m-1} and we add one more term.

5.2 Gradient boosting as functional gradient descent

Given a differentiable loss $L(y, F)$, define the empirical risk $\mathcal{L}(F) = \sum_{i=1}^n L(y_i, F(x_i))$. Functional gradient descent treats the predictions $(F(x_1), \dots, F(x_n))$ as the parameters and steps in the negative gradient direction. The negative gradient at observation i is the **pseudo-residual**

$$r_{im} = -\frac{\partial L(y_i, F(x_i))}{\partial F(x_i)} \Big|_{F=F_{m-1}}.$$

For squared-error loss $L(y, F) = \frac{1}{2}(y - F)^2$ this is exactly $r_{im} = y_i - F_{m-1}(x_i)$, the ordinary residual. At each stage we fit a base learner h_m to the pseudo-residuals (least-squares fit of $h_m(x_i)$ to r_{im}) and update

$$F_m(x) = F_{m-1}(x) + \nu h_m(x).$$

Fitting to the negative gradient is the functional analogue of a gradient descent step.

5.3 Learning rate as regularization

The learning rate $\nu \in (0, 1]$ shrinks each update. A smaller ν requires more stages M to fit the data but yields better generalization, because small steps prevent any single base learner from dominating the model. The pair (ν, M) trades off in opposite directions and is chosen by validation; small ν with large M is the standard regularizing choice.

6 Model selection

We compare candidate models fitted by maximum likelihood. Let $\hat{L}$ be the maximized likelihood, $\hat{\ell} = \log \hat{L}$ the maximized log-likelihood, k the number of free parameters, and n the sample size.

6.1 AIC

Larger models fit the training data better, so log-likelihood alone always favours complexity. The Akaike Information Criterion estimates the expected out-of-sample Kullback–Leibler divergence between the fitted model and the truth. That expected divergence equals the maximized log-likelihood minus an asymptotic bias correction equal to the number of parameters k . Defining the criterion on the deviance scale ($-2\hat{\ell}$) gives

$$\text{AIC} = -2\hat{\ell} + 2k.$$

The term $2k$ is the complexity penalty; the preferred model minimizes AIC.

6.2 BIC

The Bayesian Information Criterion arises from a Laplace approximation to the marginal likelihood $\int p(\text{data} \mid \theta)\pi(\theta) d\theta$ of each model. Taking $-2\log$ of that approximation and dropping terms that stay bounded as $n \rightarrow \infty$ leaves

$$\text{BIC} = -2\hat{\ell} + k \log n.$$

The preferred model maximizes the (approximate) posterior probability, i.e. minimizes BIC. Its Bayesian interpretation is that BIC differences approximate $-2\log$ of the Bayes factor between models.

6.3 AIC vs. BIC tradeoff

The penalties differ only in the multiplier of k : 2 for AIC, $\log n$ for BIC. For $n \geq 8$, $\log n > 2$, so BIC penalizes complexity more heavily and selects smaller models. BIC is consistent — if the true model is among the candidates, it selects it with probability tending to 1 — whereas AIC targets predictive accuracy and may keep extra parameters. The two disagree when the sample is large and a more complex model improves fit only slightly: AIC may keep it, BIC discards it.

7 Practical comparison of methods

- **Parametric (MLE).** Assume a density family and estimate its parameters. Best when the family is correct or nearly so: most efficient, fewest data needed, fully interpretable; badly biased under misspecification.
- **Non-parametric (KDE).** Assume only smoothness. Best when the shape is unknown and one-dimensional or low-dimensional with ample data; flexible but suffers the curse of dimensionality and needs more data.
- **Tree-based.** Assume nothing about functional form; handle mixed feature types, interactions, and non-linearity. Single trees are interpretable but high-variance; ensembles (random forests, gradient boosting) are accurate but less interpretable.

Choose parametric methods when a credible model exists and interpretability or efficiency is paramount; non-parametric density estimation when only the distribution’s shape is sought; tree-based ensembles when prediction accuracy on tabular data with complex structure dominates and the loss of interpretability is acceptable.

7.1 A model selection pipeline

To select a model on a given dataset: (1) propose candidate models spanning parametric, non-parametric, and tree-based families; (2) fit each, using cross-validation to set tuning parameters (bandwidth, λ , tree depth, ν and M); (3) compare nested parametric candidates by AIC/BIC

and all candidates by cross-validated predictive error; (4) document the rationale for the final choice, weighing predictive accuracy against interpretability.